# LESSON PLAN

## EWIS (21)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 08 |
| **Total No. of Lesson Plans** | 10 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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| **1. Topic** | Purpose and technical details of ADF |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Purpose and technical details of ADF. Basic layout of intercom system in aircraft. |
| **5. Specific Objectives** | 21.1.1.1 Trainee will be able to understand the basic theory of communication. |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Demonstration of ADF indicator operation and Intercom panel layout; Safety: Ensure no loose items near equipment; Supervision: Instructor monitors trainee interaction with panel. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What is the primary purpose of the ADF system?
(b) Name two types of radio navigation aids.
(c) Define the basic theory of communication.

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Basic theory of communication<br>• Purpose of ADF navigation<br>• ADF technical details and components<br>• ADF signal reception principles<br>• ADF indicator operation | L & Ds | PPT, IB | 15 min |
| (b) | • Basic layout of intercom system<br>• Intercom panel components<br>• Audio signal routing paths<br>• Microphone and speaker locations<br>• Intercom switch functions | L & Ds | PPT, PDF | 20 min |
| (c) | • ADF and Intercom integration<br>• Troubleshooting common faults<br>• Safety checks for radio equipment<br>• Operational procedures review<br>• Maintenance requirements overview | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain how the ADF determines aircraft position relative to a station.
(b) Describe the function of the intercom panel in the aircraft cabin.

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| **12. Home Assignment and Follow Up** | Prepare a brief note on the differences between ADF and VOR systems to be discussed in LP 09. |

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| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Purpose and function of Altimeter and their types. Global positioning system (GPS) introduction, purpose airborne and space components of the system. Safety precautions to be observed whilst working on avionics systems | Refer to the master syllabus reference list | LP 09 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min